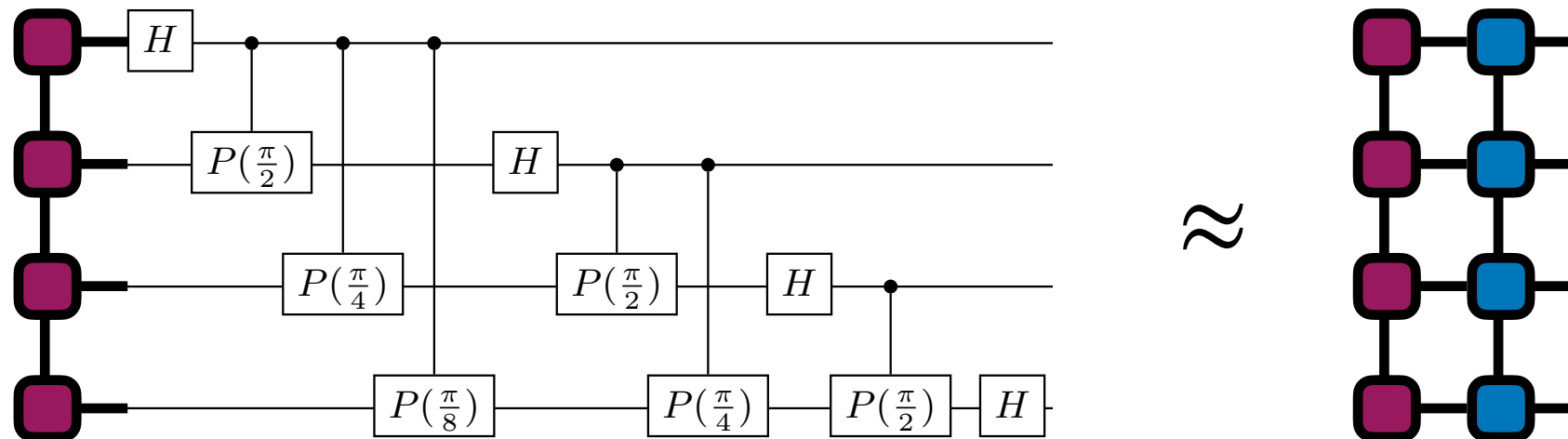


Time Evolution with Tensor Networks



Tensor Network Algorithms

An important capability is
time evolving tensor networks

$$e^{-iHt} |\psi(0)\rangle = |\psi(t)\rangle$$

Algorithms:

- Time evolving block decimation (TEBD)
- Time-dependent variational principle (TDVP)

Tensor Network Algorithms

TEBD algorithm $H = \sum_j h_{j,j+1}$

Tensor Network Algorithms

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$$U = e^{-iHt} = \prod_{k=1}^n e^{-iH\delta t} \quad n\delta t = t$$

Tensor Network Algorithms

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Want to decompose $e^{-iH\delta t}$ to keep Trotter Error low in a given time step. Use Suzuki-Trotter

Tensor Network Algorithms

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1st Order: $e^{-iH\delta t} = \prod_i e^{-ih_{i,i+1}\delta t} + \mathcal{O}(\delta t^2)$

Tensor Network Algorithms

TEBD algorithm $H = \sum_j h_{j,j+1}$

$$U = e^{-iHt} = \prod_{k=1}^n e^{-iH\delta t} \quad n\delta t = t$$

Want to decompose $e^{-iH\delta t}$ to keep Trotter Error low in a given time step. Use Suzuki-Trotter

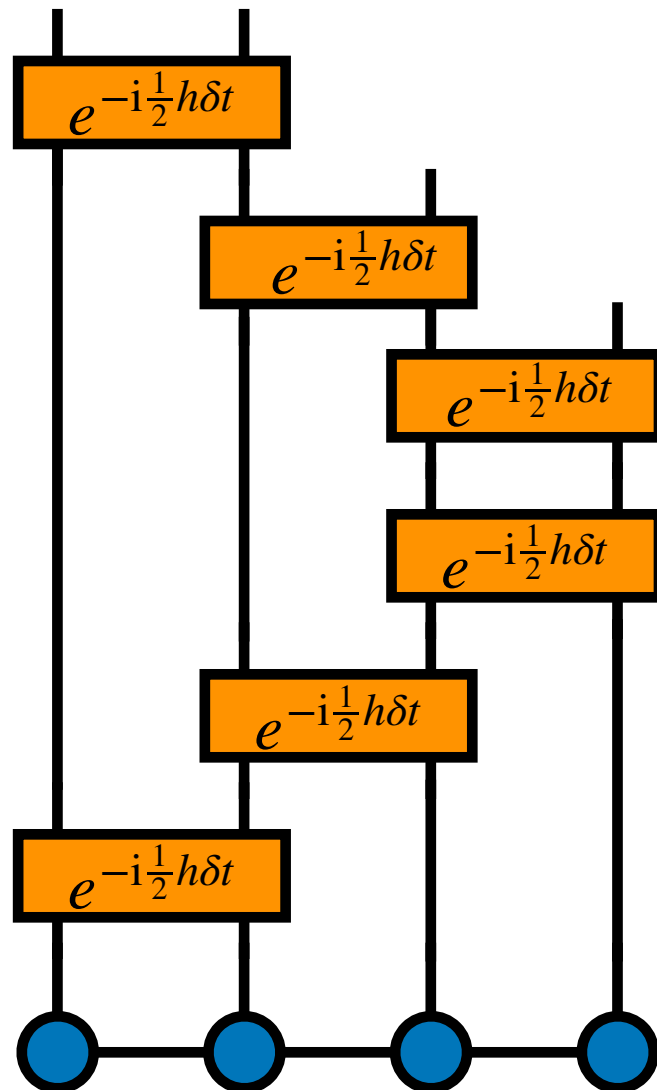
1st Order: $e^{-iH\delta t} = \prod_i e^{-ih_{i,i+1}\delta t} + \mathcal{O}(\delta t^2)$

2nd Order: $e^{-iH\delta t} = \left(\prod_{1 \leq i \leq N} e^{-\frac{1}{2}ih_{i,i+1}\delta t} \right) \left(\prod_{N \geq i \geq 1} e^{-\frac{1}{2}ih_{i,i+1}\delta t} \right) + \mathcal{O}(\delta t^3)$

Tensor Network Algorithms

TEBD algorithm

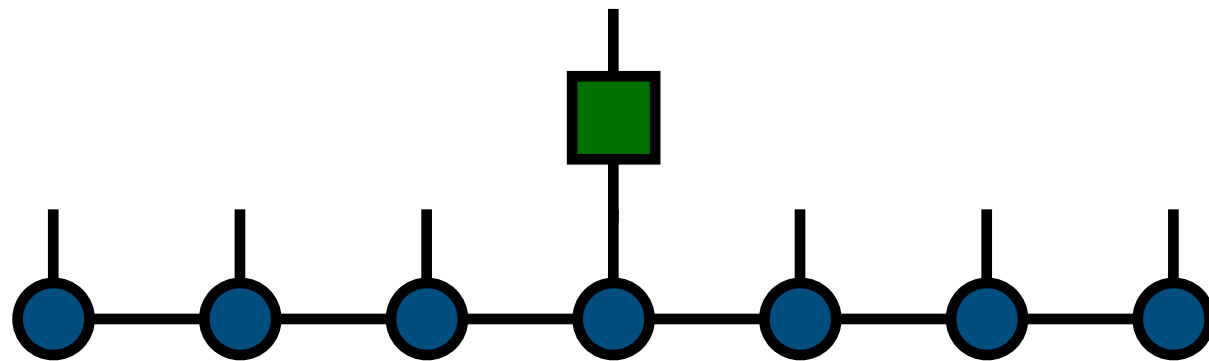
$$e^{-iH\delta t} = \left(\prod_{1 \leq i \leq N} e^{-\frac{1}{2}i h_{i,i+1} \delta t} \right) \left(\prod_{N \geq i \geq 1} e^{-\frac{1}{2}i h_{i,i+1} \delta t} \right) + \mathcal{O}(\delta t^3)$$



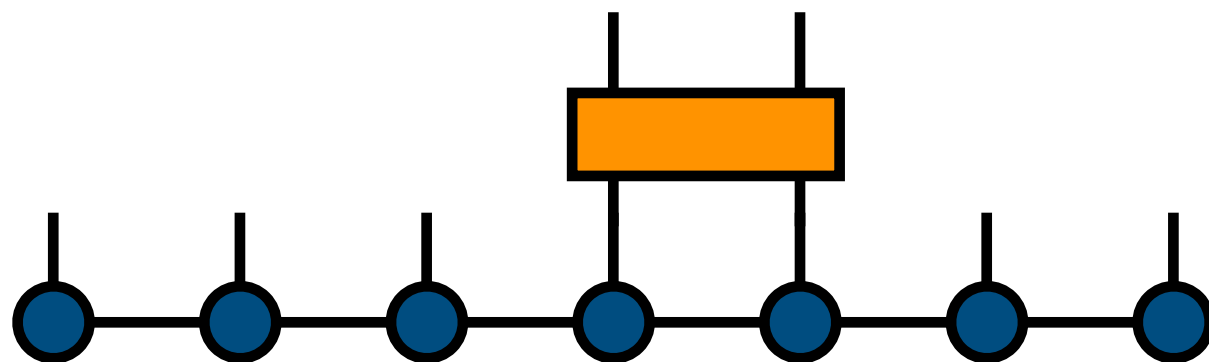
$$\approx e^{-iH\delta t} |\psi(t)\rangle = |\psi(t + \delta t)\rangle$$

Tensor Network Algorithms

TEBD = controlled application of one-qubit and two-qubit gates to an MPS



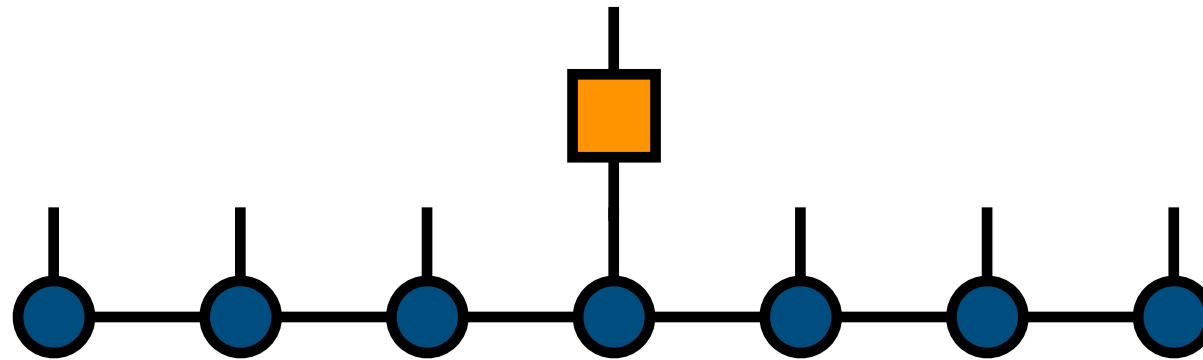
one-qubit gate is exact



two-qubit gate incurs small error controlled by bond-dimension χ

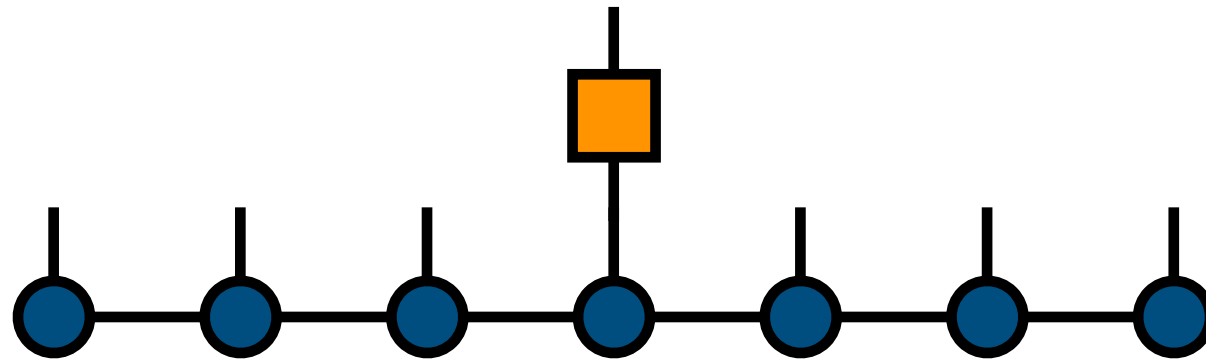
Tensor Network Algorithms

Say we want to act a single-qubit gate on a wavefunction in MPS form

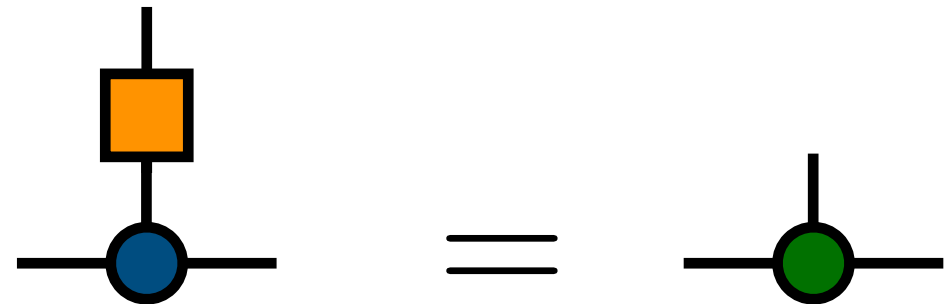


Tensor Network Algorithms

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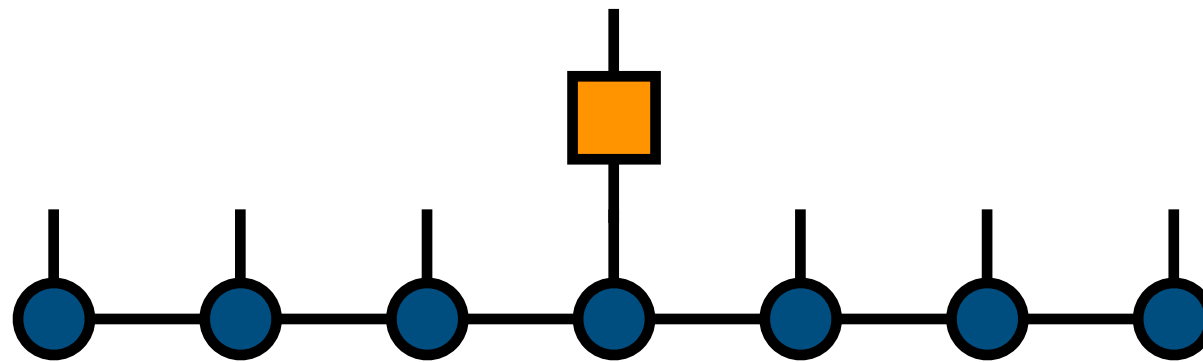


Just operate on one tensor:

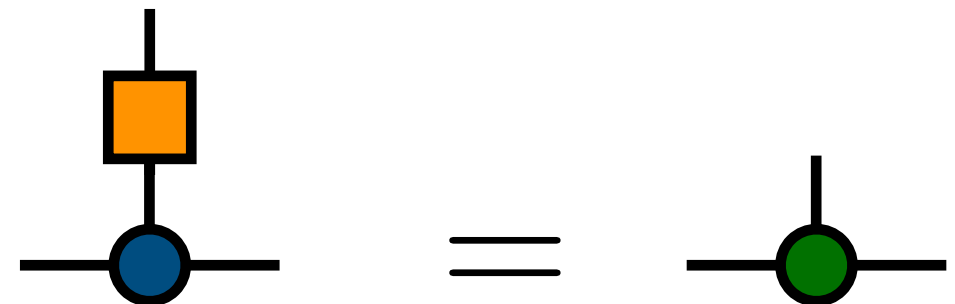


Tensor Network Algorithms

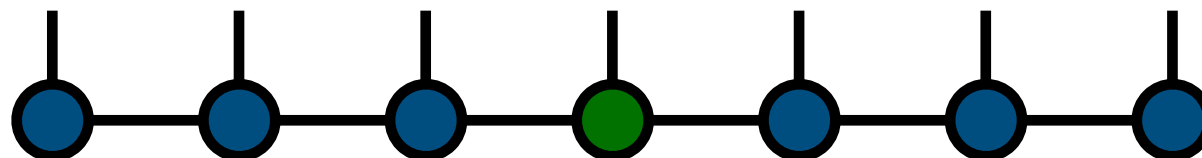
Say we want to act a single-qubit gate on a wavefunction in MPS form



Just operate on one tensor:

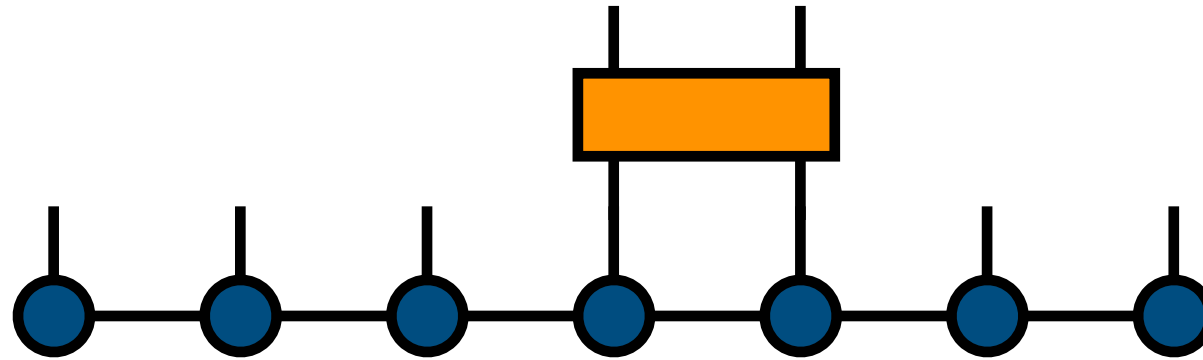


Result:

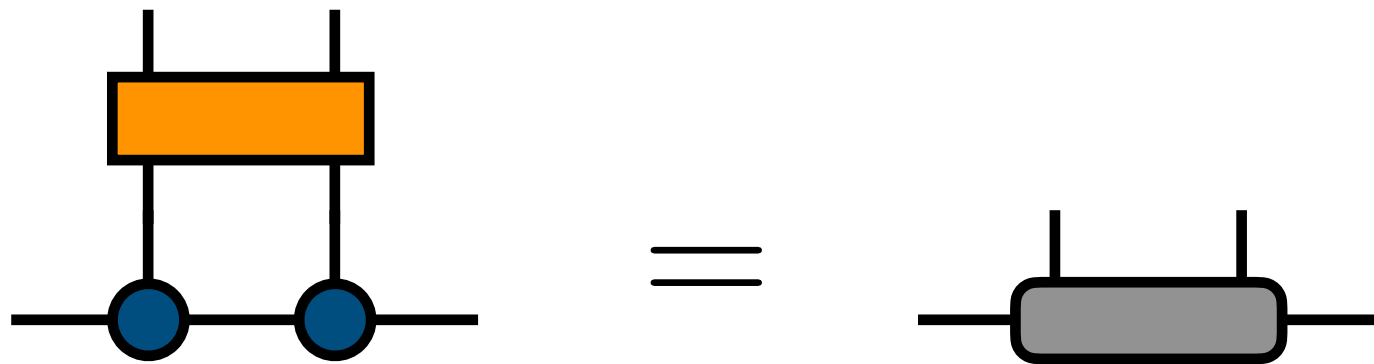


*same bond
dimension*

Say we want to act two-qubit gate on a wavefunction in MPS form

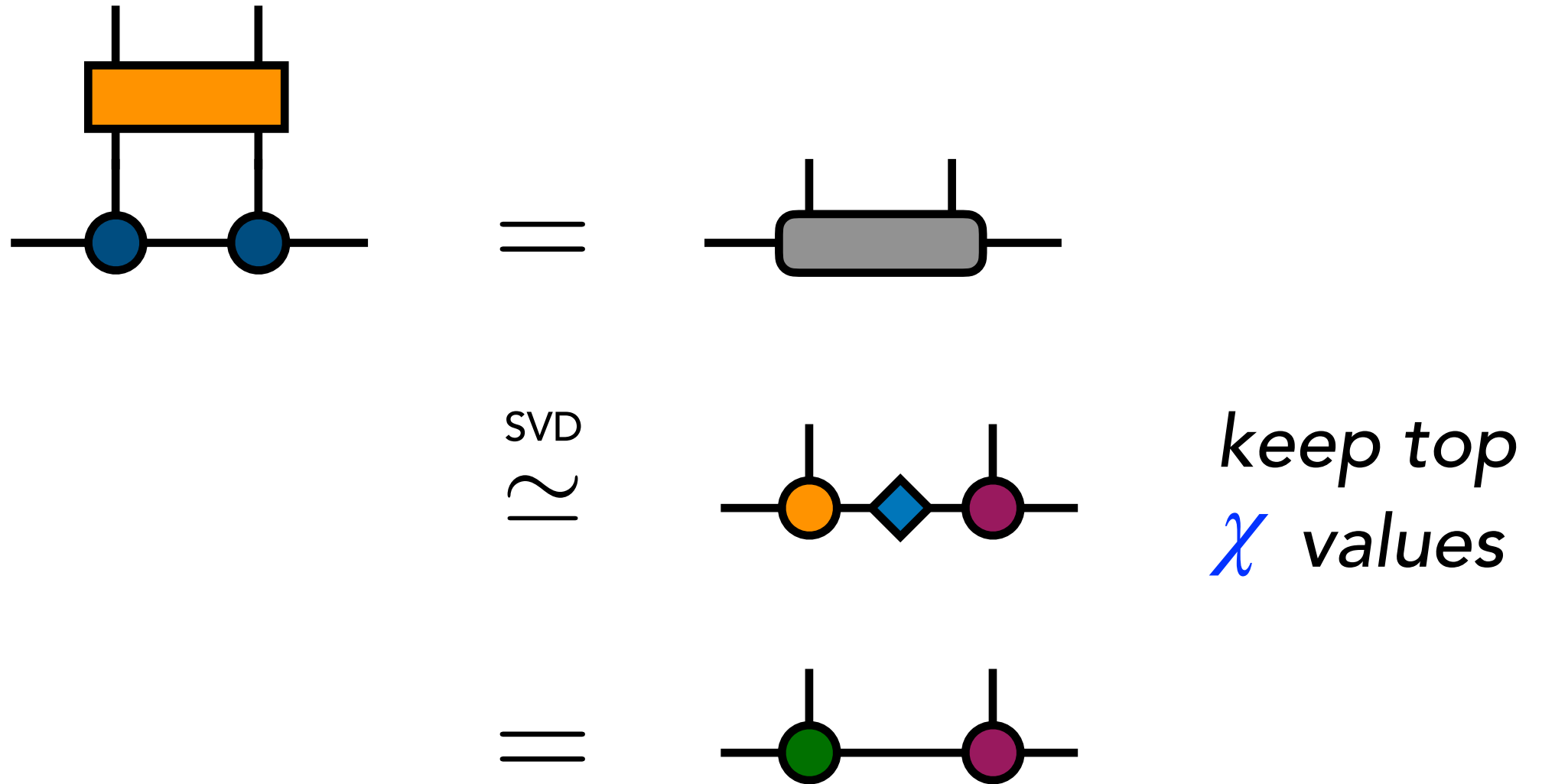


Operate on two tensors:

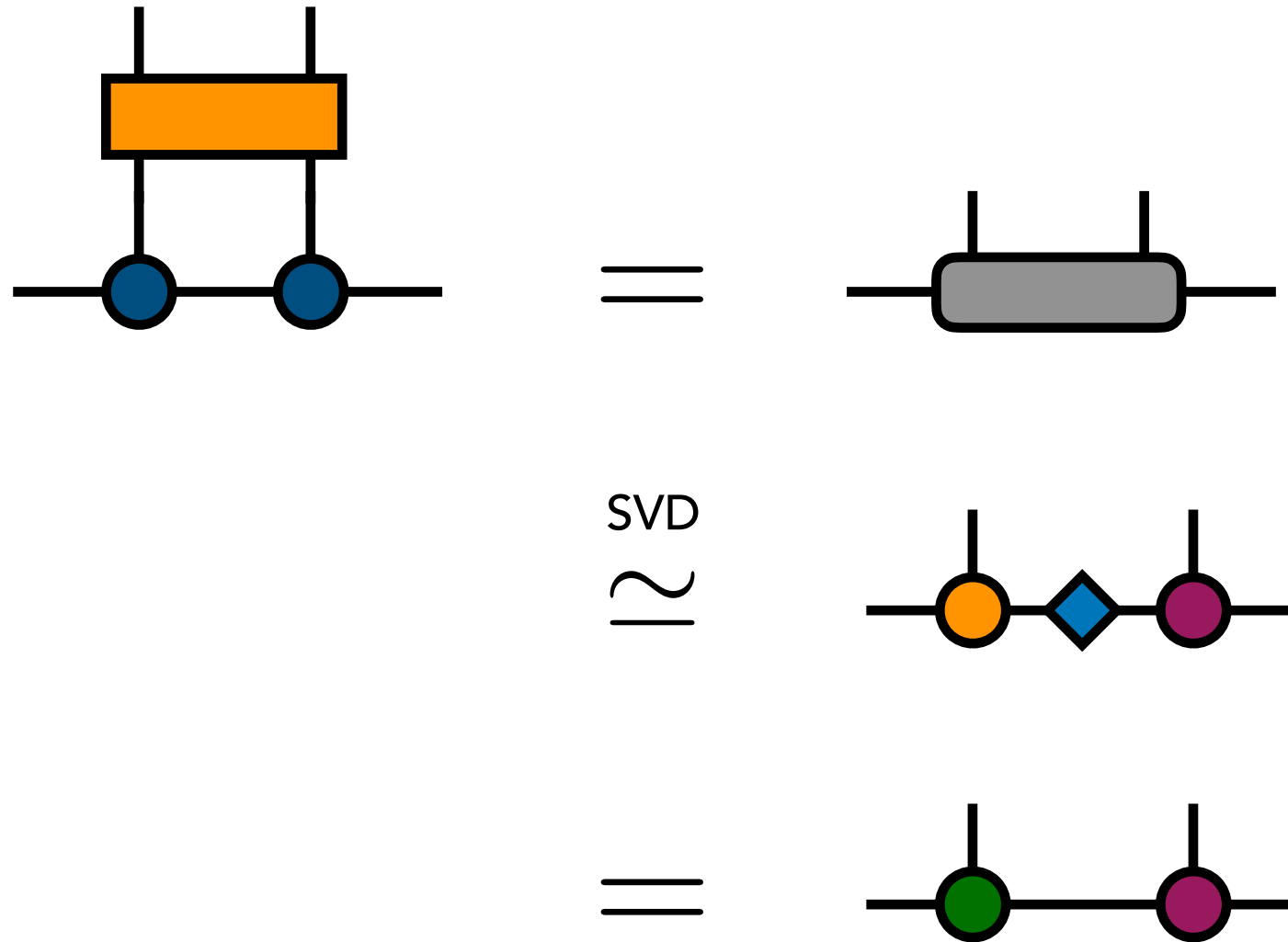


*destroy
MPS form
locally*

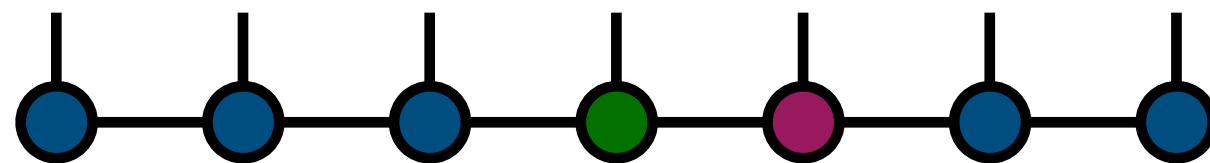
But can recover MPS form using truncated SVD:



But can recover MPS form using truncated SVD:



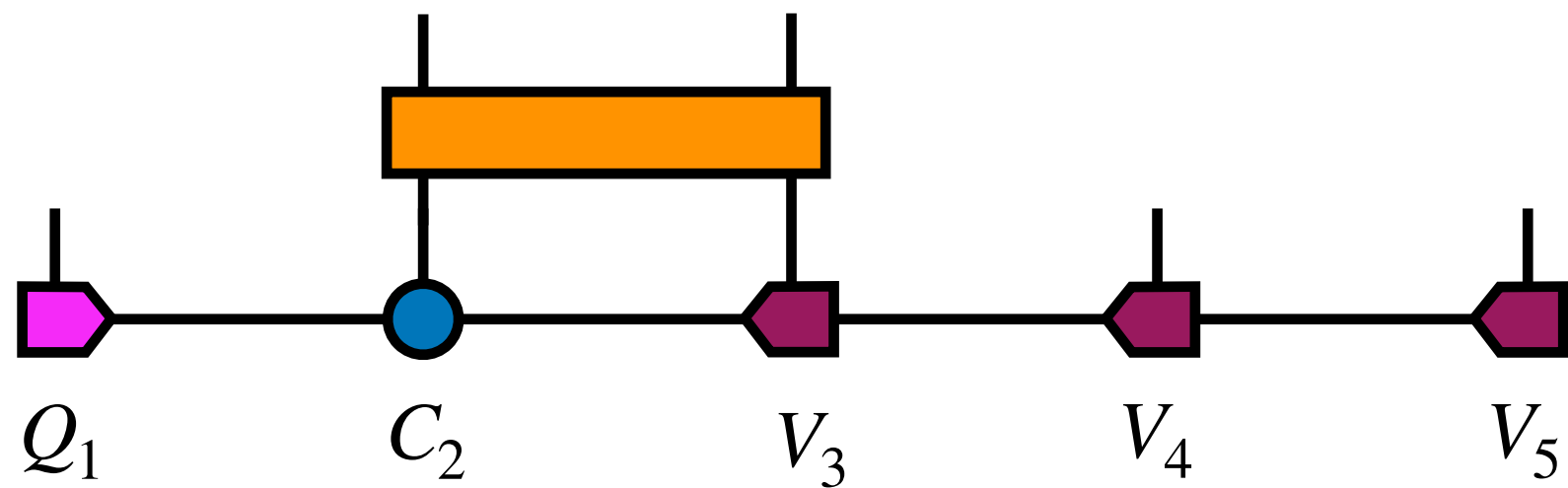
Result:



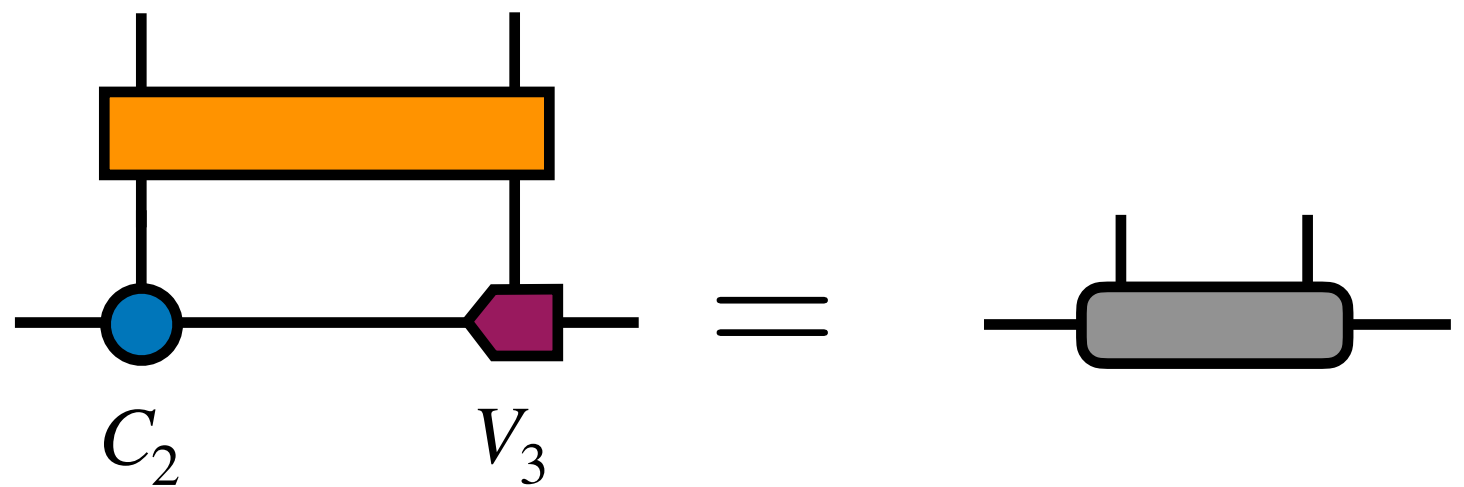
*with loss of
fidelity
dependent
on level of
truncation*

Using the Orthogonal MPS Gauge in TEBD

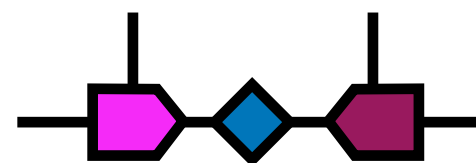
It's important to apply the gate where the gauge centre is



Can SVD and move the orthogonality centre along at the same time

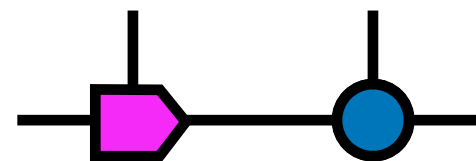


SVD
 $\sim$



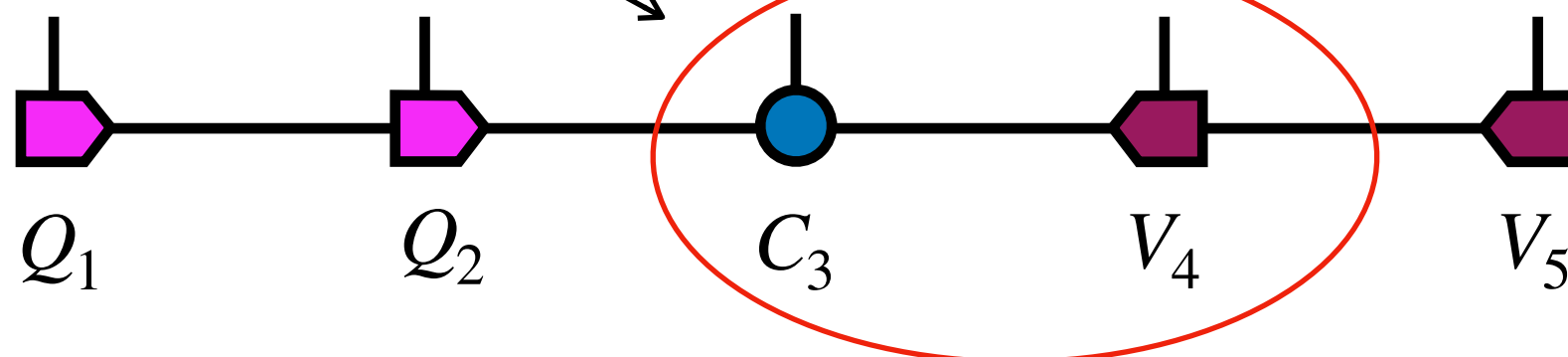
keep top
 χ values

$\equiv$



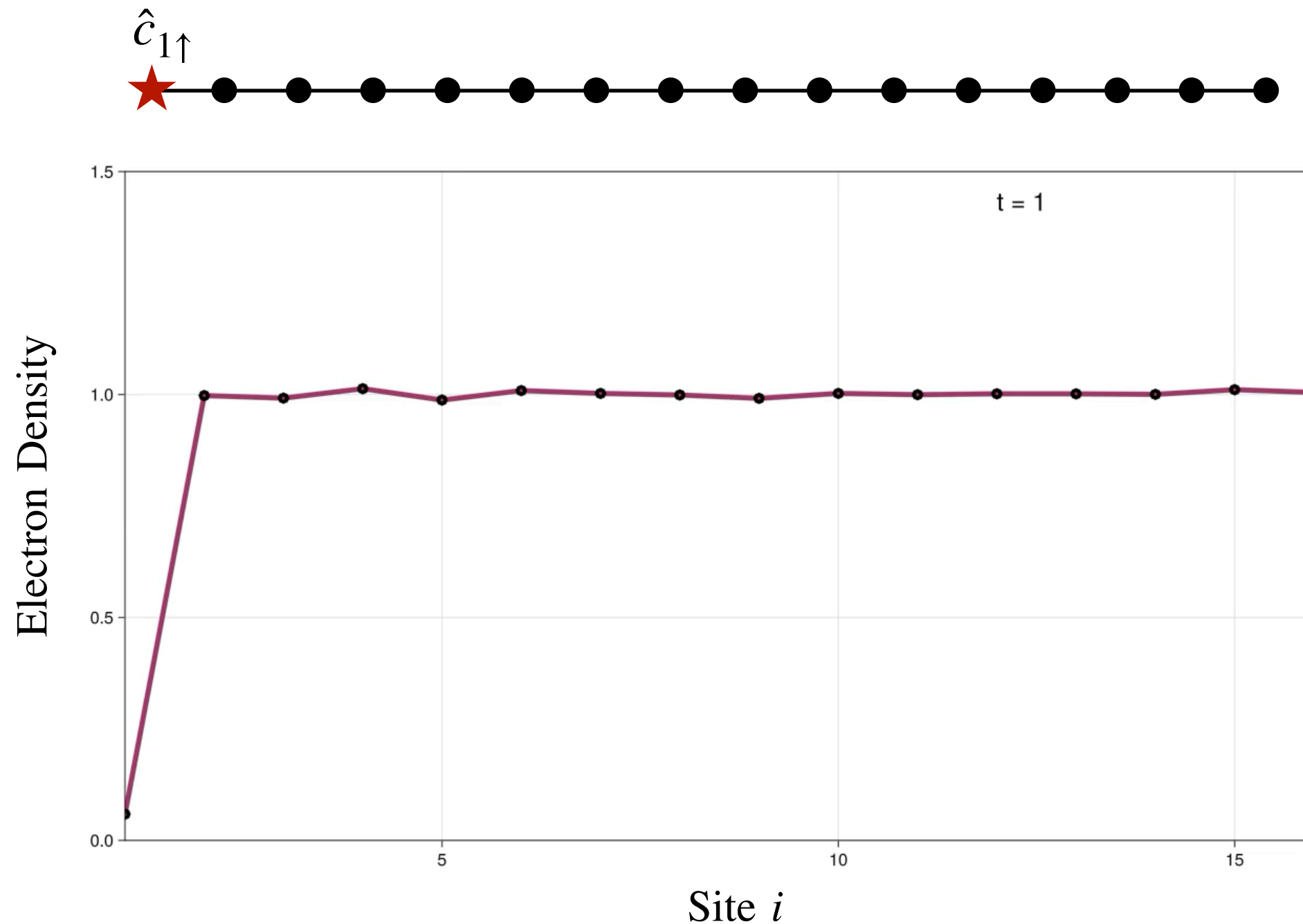
Ready for a gate!

Result:



Tensor Network Algorithms

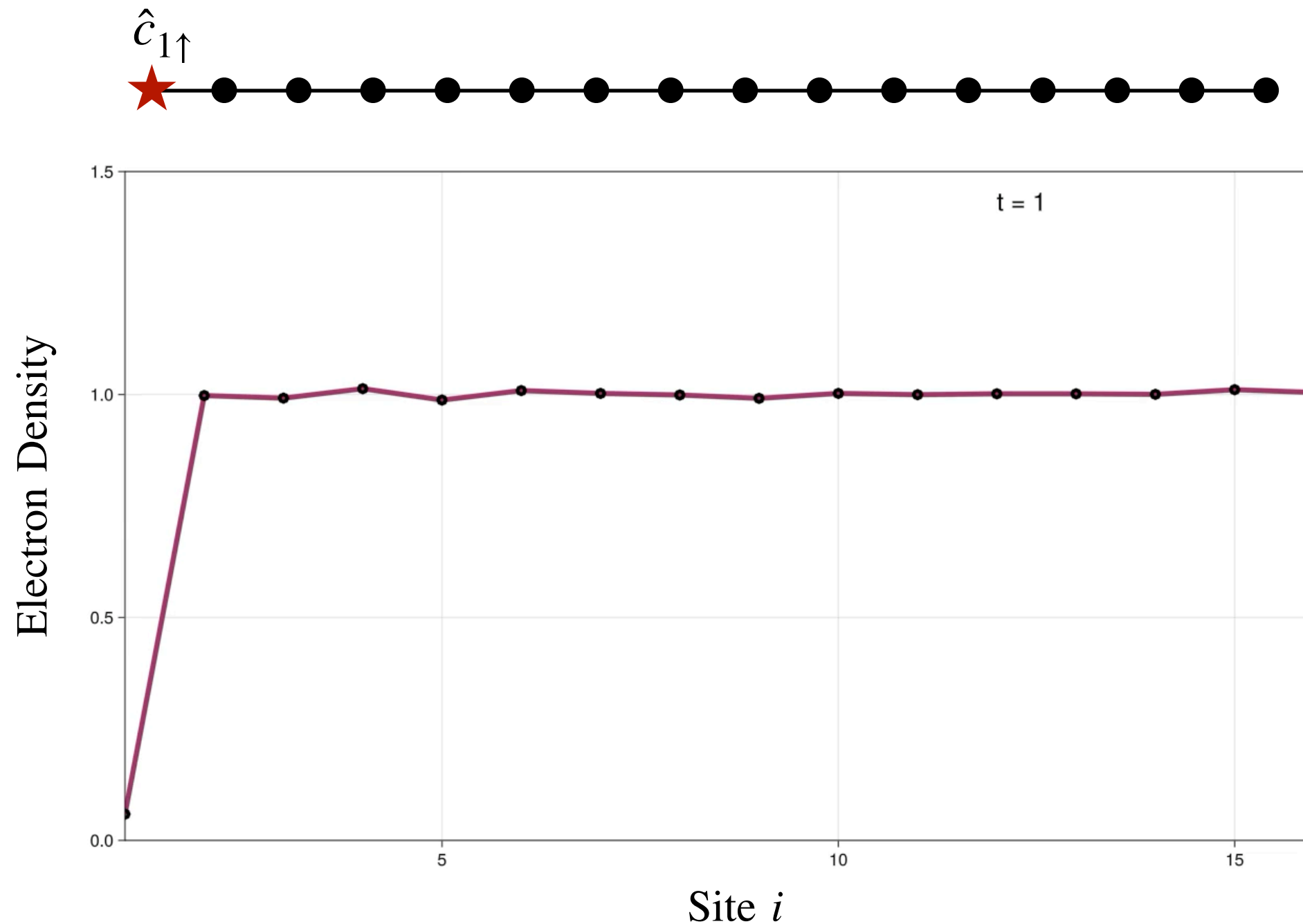
TEBD results: electron chain* after removing particle on site 1



*Impurity model with uniform hopping and $U=6$

Tensor Network Algorithms

TEBD results: electron chain* after removing particle on site 1



*Impurity model with uniform hopping and $U=6$

Tensor Network Algorithms

TEBD Limitations:

Tensor Network Algorithms

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- Entanglement of states generically **grows linearly** in time

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Time-dependent variational principle (TDVP)

Tensor Network Algorithms

TEBD Limitations:

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Time-dependent variational principle (TDVP)

- Lets us handle long range interactions

Tensor Network Algorithms

TDVP algorithm

Time evolve via a DMRG-like sweeping procedure

Start with the full Hamiltonian

$$H = \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet$$

Can be a local or non-local Hamiltonian now

Tensor Network Algorithms

TDVP algorithm [1]

$$H = \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \bullet$$

Solve the Schrödinger equation in the manifold of MPS of fixed bond dimension χ

$$\frac{d|\psi(t)\rangle}{dt} = -i\mathcal{P}_{|\psi(t)\rangle}H|\psi(t)\rangle$$

$\mathcal{P}_{|\psi(t)\rangle}$ is the projector onto the tangent space of this manifold for the current state

[1] J. Haegeman et al, TDVP for Quantum Lattices, PRL 107 (2011)

Tensor Network Algorithms

TDVP algorithm

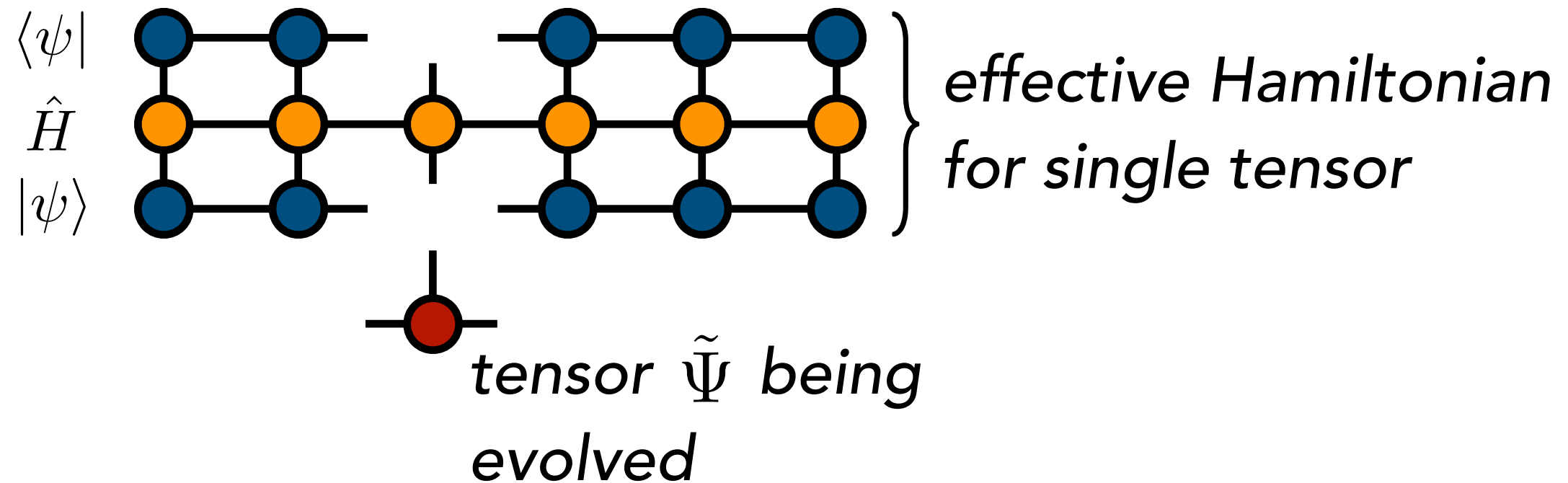
$$\frac{d|\psi(t)\rangle}{dt} = -i\mathcal{P}_{|\psi(t)\rangle}H|\psi(t)\rangle$$

$$|\psi(t + \delta t)\rangle = e^{-i\mathcal{P}_{|\psi(t)\rangle}H\delta t}|\psi(t)\rangle$$

$\mathcal{P}_{|\psi(t)\rangle}$ can be constructed exactly and we Trotterize the exponential at the level of the projector, not the Hamiltonian, solving the local integral equation that arises site by site.

Tensor Network Algorithms

TDVP algorithm



Tensor Network Algorithms

At each configuration, integrate time forward by small amount

Solving:

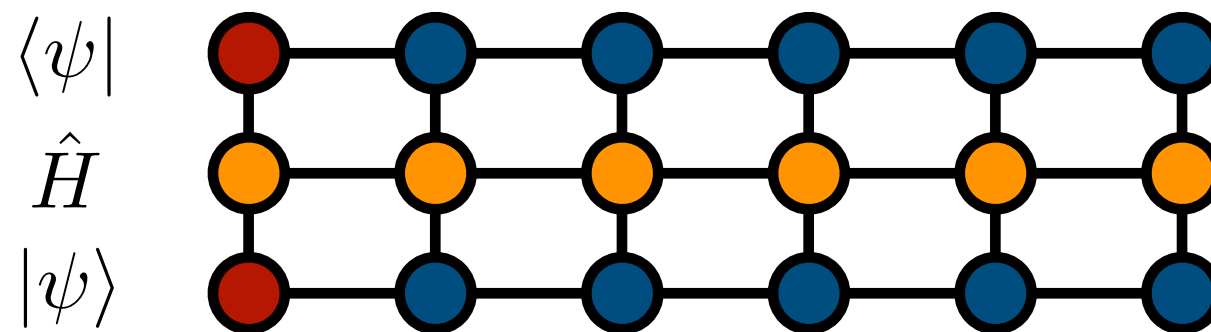
$$i \frac{d}{dt} \text{---} \textcircled{\text{red}} \text{---} = \text{---} \begin{array}{ccccc} \textcircled{\text{blue}} & \textcircled{\text{blue}} & & \textcircled{\text{blue}} & \textcircled{\text{blue}} & \textcircled{\text{blue}} \\ | & | & & | & | & | \\ \textcircled{\text{orange}} & \textcircled{\text{orange}} & \textcircled{\text{orange}} & \textcircled{\text{orange}} & \textcircled{\text{orange}} & \textcircled{\text{orange}} \\ | & | & & | & | & | \\ \textcircled{\text{blue}} & \textcircled{\text{blue}} & & \textcircled{\text{blue}} & \textcircled{\text{blue}} & \textcircled{\text{blue}} \end{array} \text{---} \textcircled{\text{red}} \text{---}$$

$$\text{from } t \rightarrow t + dt = \text{---} \left(\begin{array}{c} \text{---} \textcircled{\text{blue}} \text{---} \\ | \\ \text{---} \textcircled{\text{orange}} \text{---} \\ | \\ \text{---} \textcircled{\text{blue}} \text{---} \end{array} \right) \hat{H}_{\text{Eff}} \left(\begin{array}{c} \text{---} \textcircled{\text{blue}} \text{---} \\ | \\ \text{---} \textcircled{\text{orange}} \text{---} \\ | \\ \text{---} \textcircled{\text{blue}} \text{---} \end{array} \right) \text{---} \textcircled{\text{red}} \text{---} = \hat{H}_{\text{Eff}} \cdot \text{---} \textcircled{\text{red}} \text{---}$$

Tensor Network Algorithms

TDVP algorithm

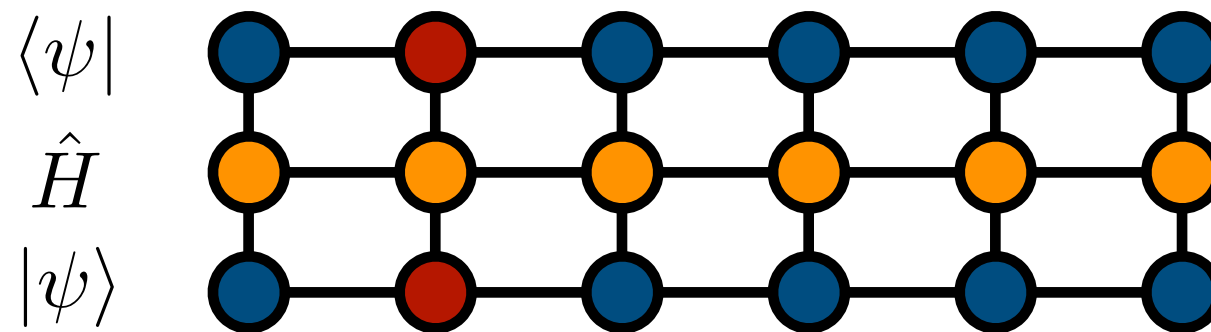
“Sweep” just like in DMRG, but solve a different local optimization problem (by integrating the local equation of motion induced by the Projected Hamiltonian $e^{-i\mathcal{P}_{|\psi(t)\rangle}H\delta t}$)



Tensor Network Algorithms

TDVP algorithm

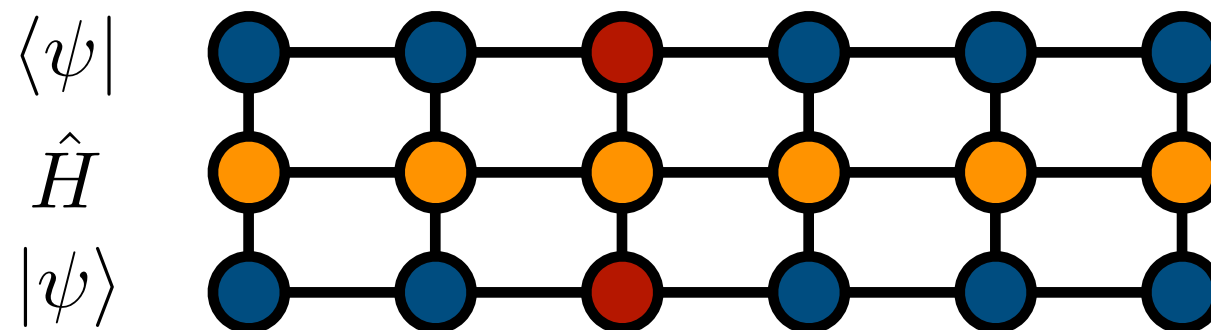
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Tensor Network Algorithms

TDVP algorithm

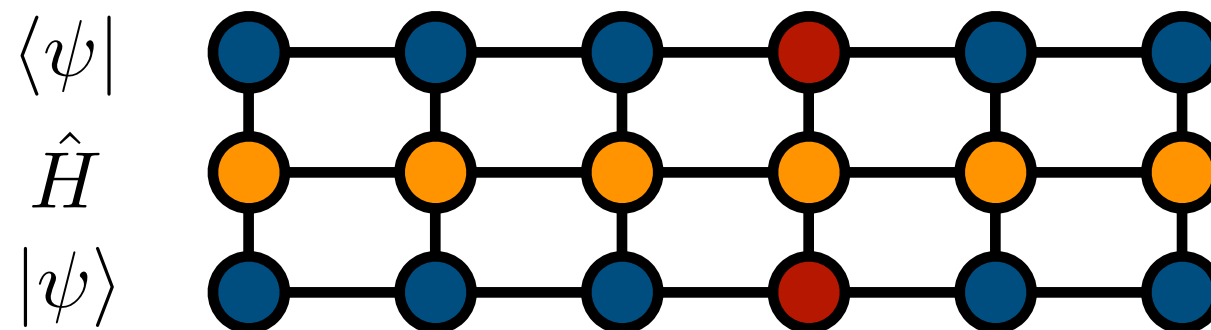
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Tensor Network Algorithms

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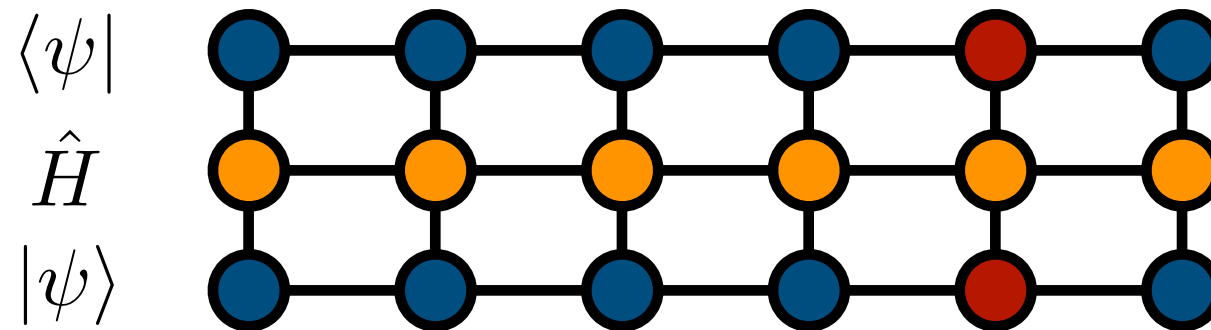
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Tensor Network Algorithms

TDVP algorithm

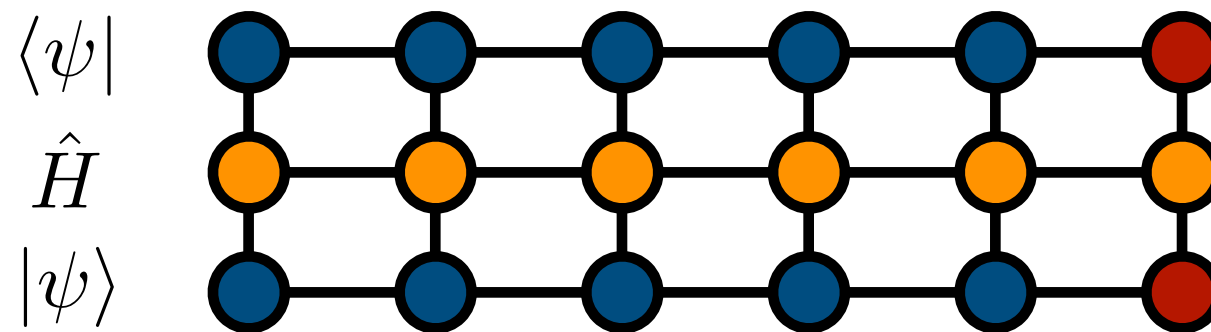
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Tensor Network Algorithms

TDVP algorithm

“Sweep” just like in DMRG, but solve a different local optimization problem (by integrating the local equation of motion induced by the Projected Hamiltonian $e^{-i\mathcal{P}_{|\psi(t)\rangle}H\delta t}$)



Tensor Network Algorithms

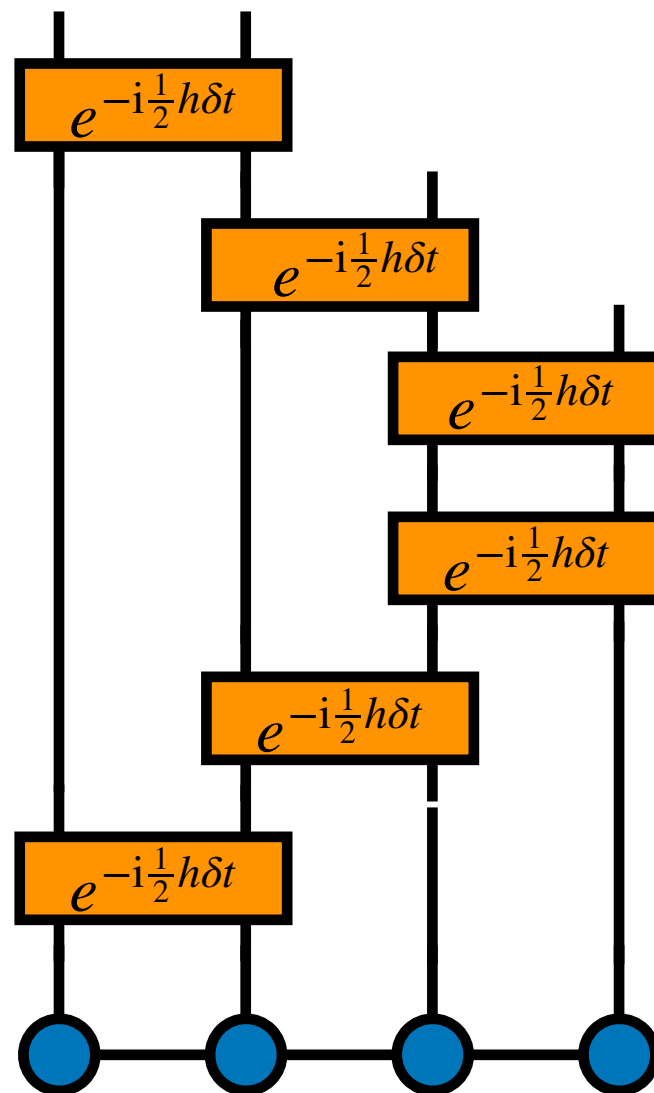
TDVP algorithm

- Sweeping algorithm, like DMRG. One-site version (fixed bond dimension) and two-site (adaptive bond dimension) versions exist.
- Scales as $\mathcal{O}(L\chi^3 t)$ for a fixed total time t , just like TEBD.
- Handles long-range interactions

TDVP and TEBD Applications

What can we do with these time-evolution algorithms?

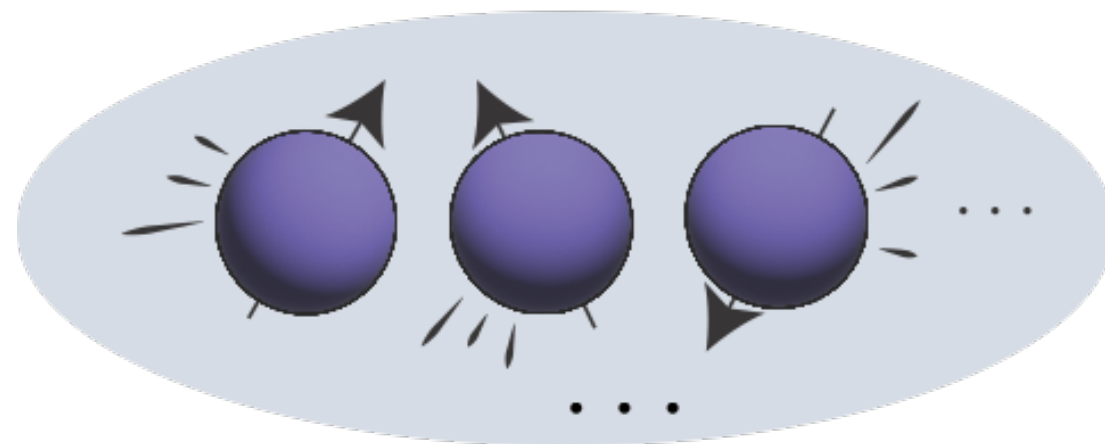
- Time evolving block decimation (TEBD)
- Time-dependent variational principle (TDVP)



TDVP and TEBD Applications

Quenches are the immediate use case of TEBD and TDVP

$$e^{-iHt} |\psi(0)\rangle = |\psi(t)\rangle$$

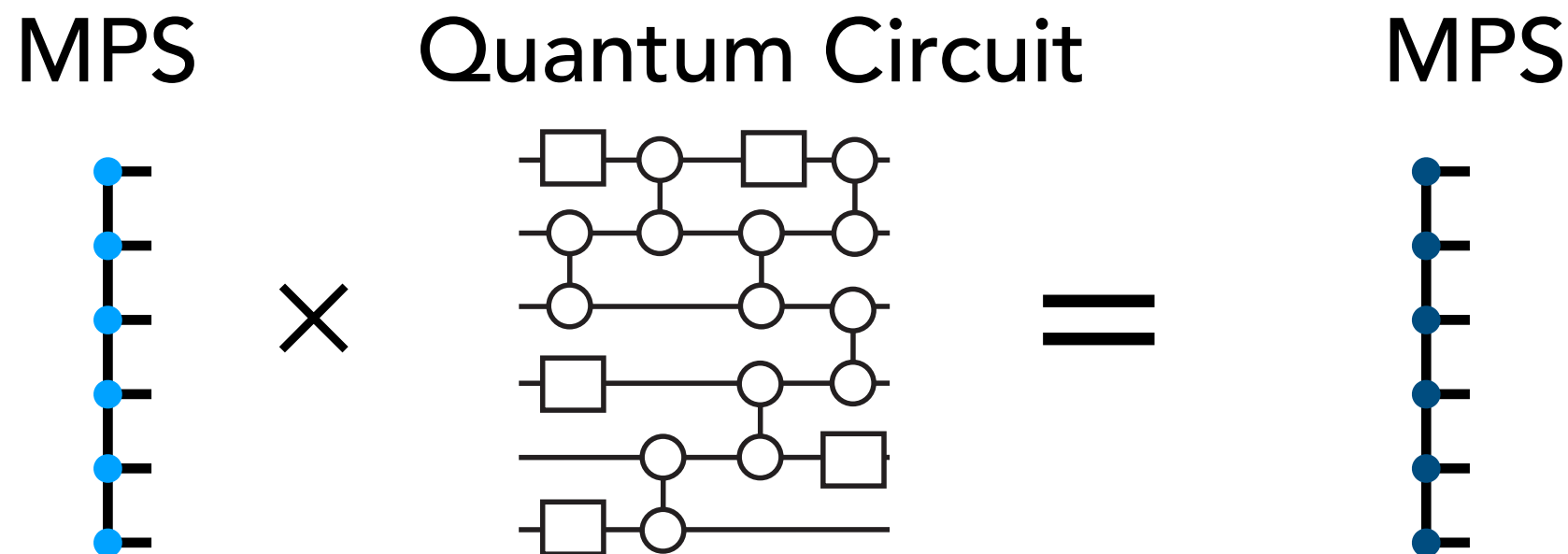


Algorithm Runtime: $\mathcal{O}(L\chi^3 t)$

Out-of-equilibrium physics is a very rich,
active research topic

TDVP and TEBD Applications

Simulating quantum circuits is possible with TEBD

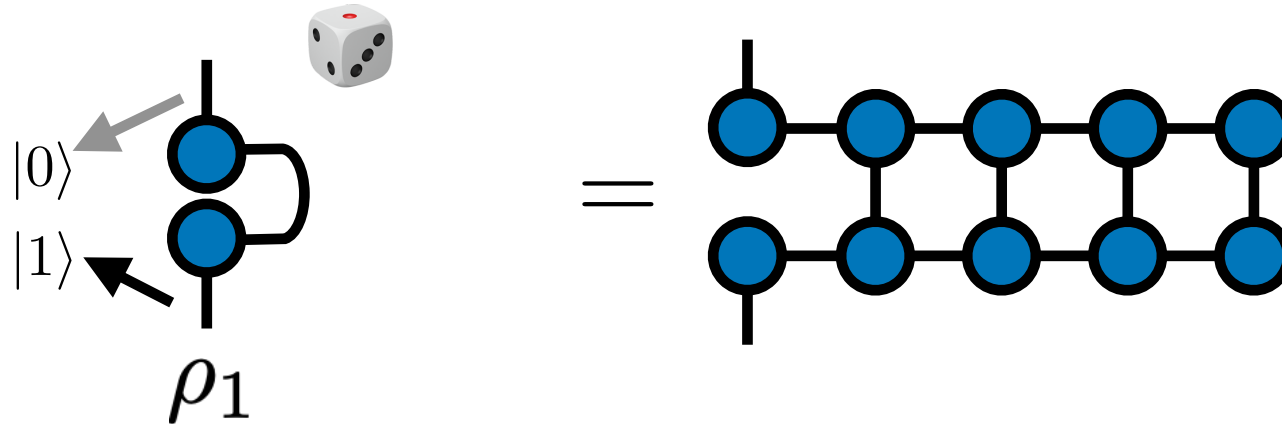


Can even sample perfectly $x \sim P(x) = |\langle x | \psi \rangle|^2$ from an MPS $|\psi\rangle$ in $\mathcal{O}(L\chi^3)$ time

- Zhou, EMS, Waintal, "What Limits the Simulation of Quantum Computers?", [PRX 10, 041038](#) (2020)
- Ayral et al., "DMRG Algorithm for Simulating Quantum Circuits with a Finite Fidelity", [PRX Quantum 4, 020304](#) (2023)

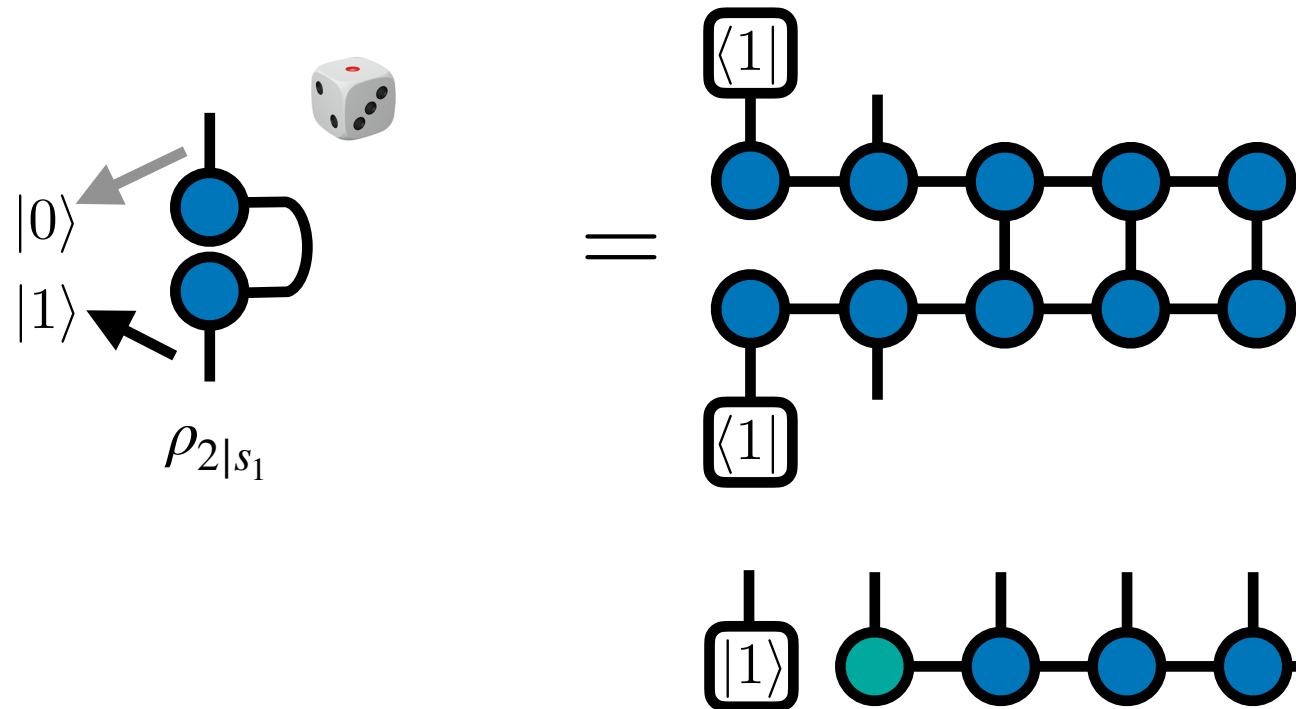
A Brief Aside on Sampling

Can "perfect sample" MPS tensor networks
No Markov chain / autocorrelation effects



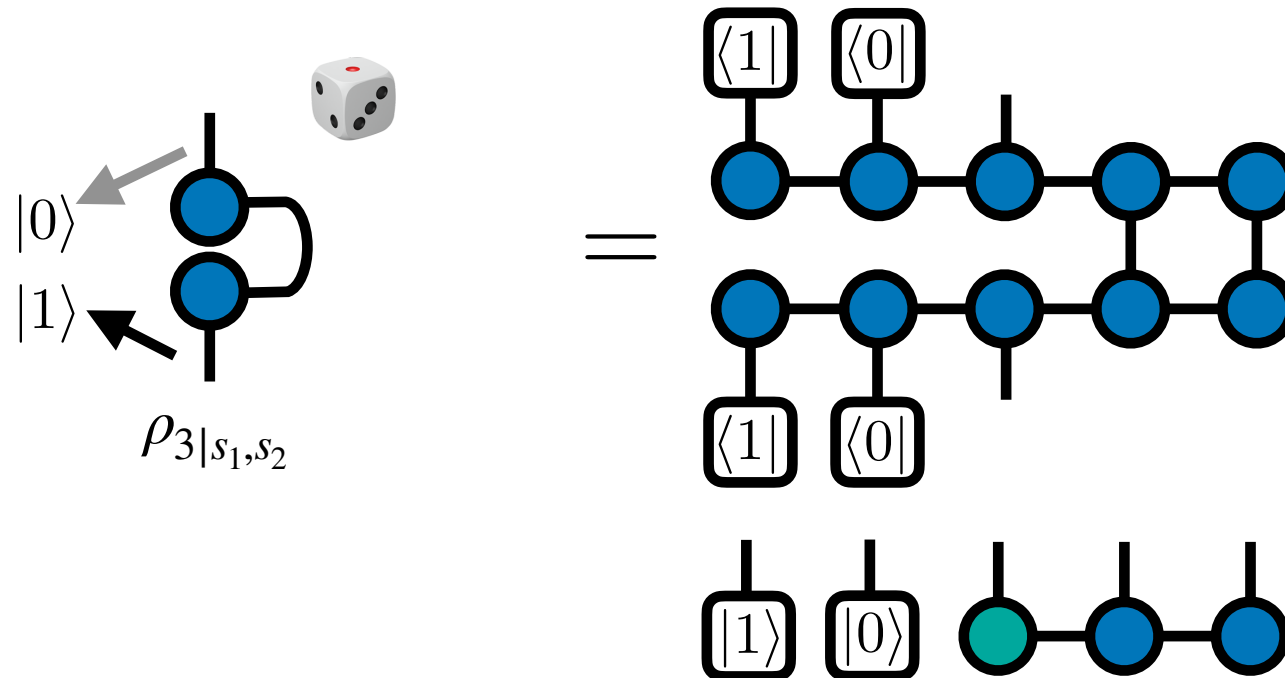
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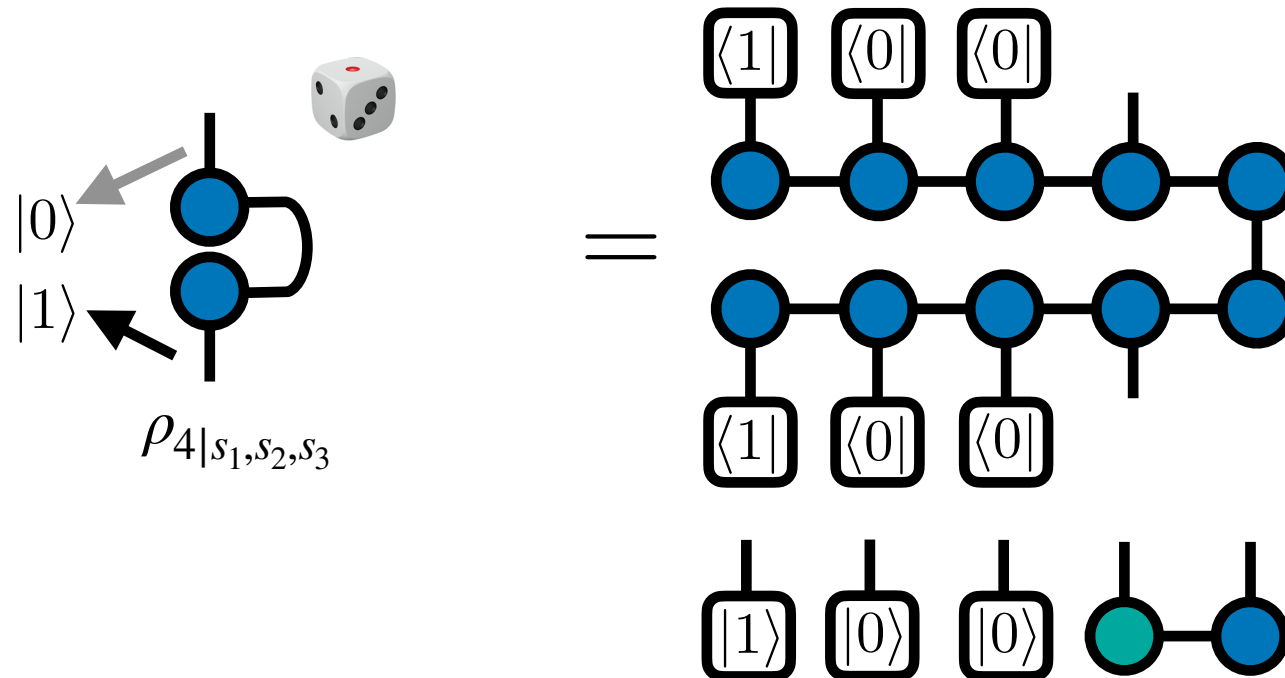
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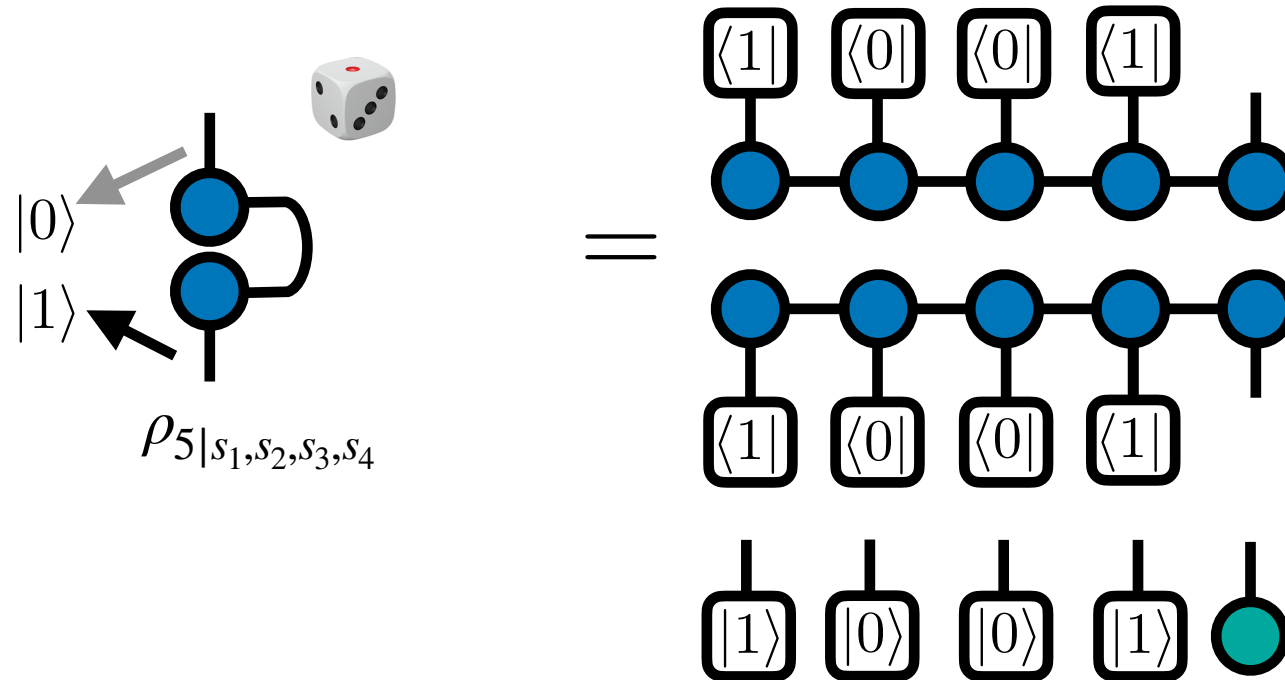
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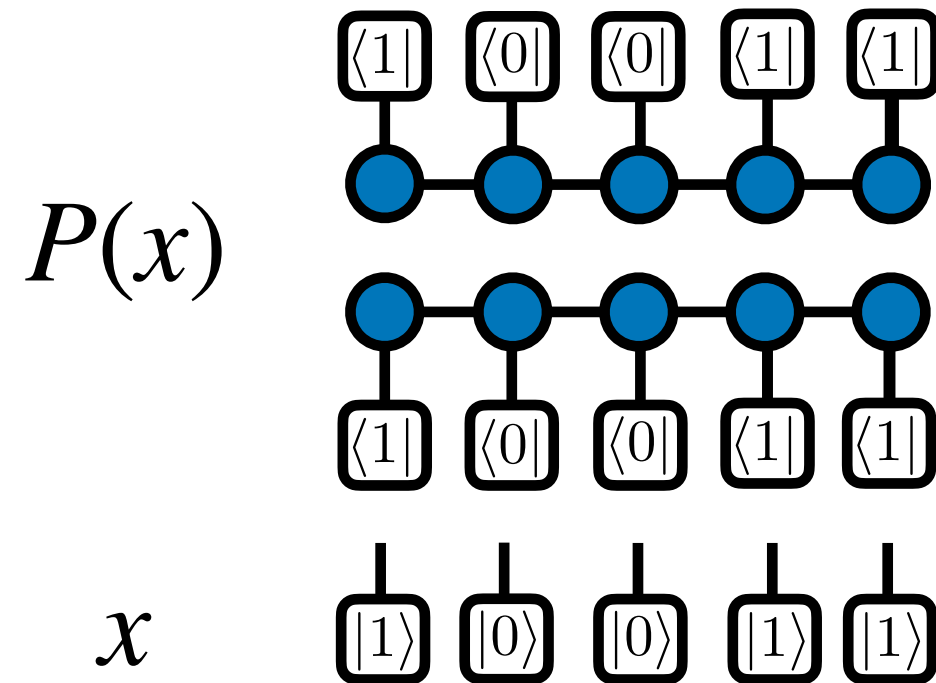
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A Brief Aside on Sampling

Now we have our sample $x \sim P(x) = |\langle x | \psi \rangle|^2$

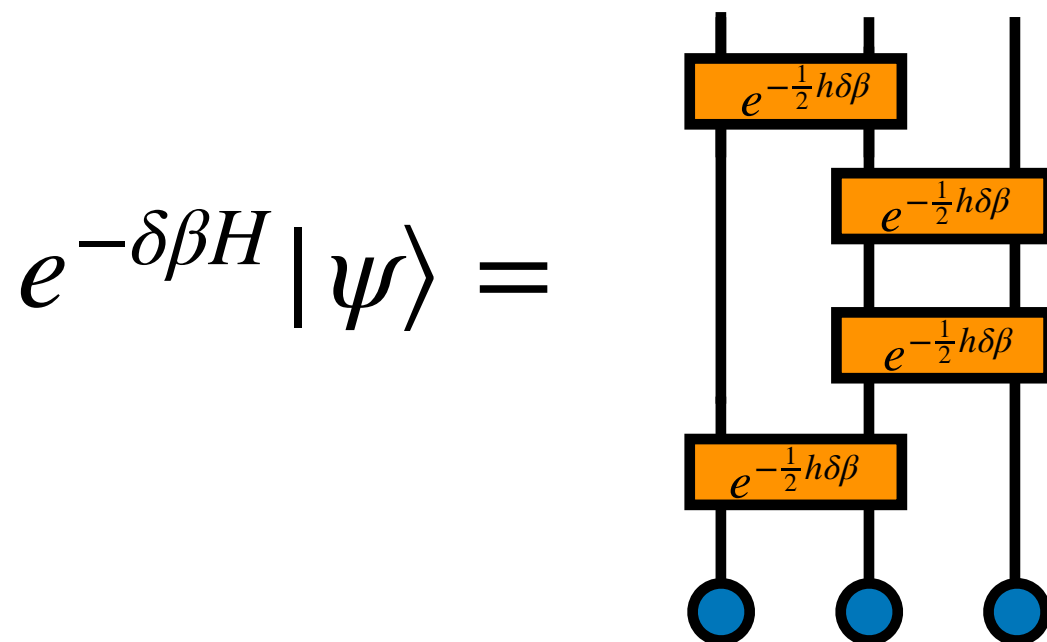


TDVP and TEBD Applications

Imaginary time evolution (TEBD or TDVP)

We can evolve in imaginary time with $t \rightarrow i\beta$ to get ground states or even thermal states (by evolving an MPO)

$$\lim_{\beta \rightarrow \infty} e^{-\beta H} |\psi\rangle = \lim_{n \rightarrow \infty} \prod_{k=1}^n e^{-\delta\beta H} |\psi\rangle \rightarrow |\psi_{GS}\rangle$$



TDVP and TEBD Applications

We can even do finite temperature whilst staying in the pure state picture

First observe, for any basis, $\beta = \frac{1}{T}$

$$e^{-\beta \hat{H}} = \sum_i \underbrace{e^{-\frac{\beta}{2} \hat{H}} |i\rangle \langle i| e^{-\frac{\beta}{2} \hat{H}}}_{\text{Freedom to choose these}} \propto \sum_i |\phi_i\rangle \langle \phi_i|$$

Freedom to
choose these $|\phi_i\rangle \propto e^{-\frac{\beta}{2} \hat{H}} |i\rangle$

We have

$$e^{-\beta \hat{H}} = \sum_i e^{-\frac{\beta}{2} \hat{H}} |i\rangle \langle i| e^{-\frac{\beta}{2} \hat{H}} \propto \sum_i |\phi_i\rangle \langle \phi_i|$$

Obtain observables as

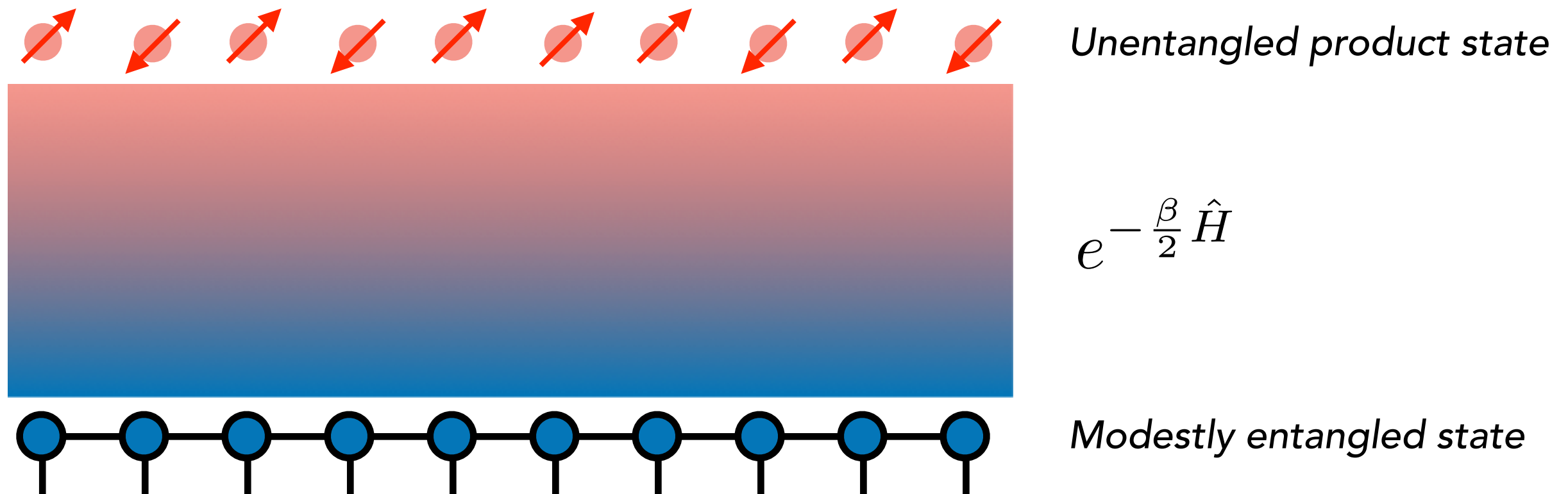
$$\langle \hat{A} \rangle = \frac{1}{\mathcal{Z}} \text{Tr} [e^{-\beta \hat{H}} \hat{A}] = \frac{1}{\mathcal{Z}} \sum_i P_i \langle \phi_i | \hat{A} | \phi_i \rangle$$

↑
an average over
pure states

Expanding $e^{-\beta \hat{H}} \propto \sum_i |\phi_i\rangle \langle \phi_i|$

To give tensor networks their best chance

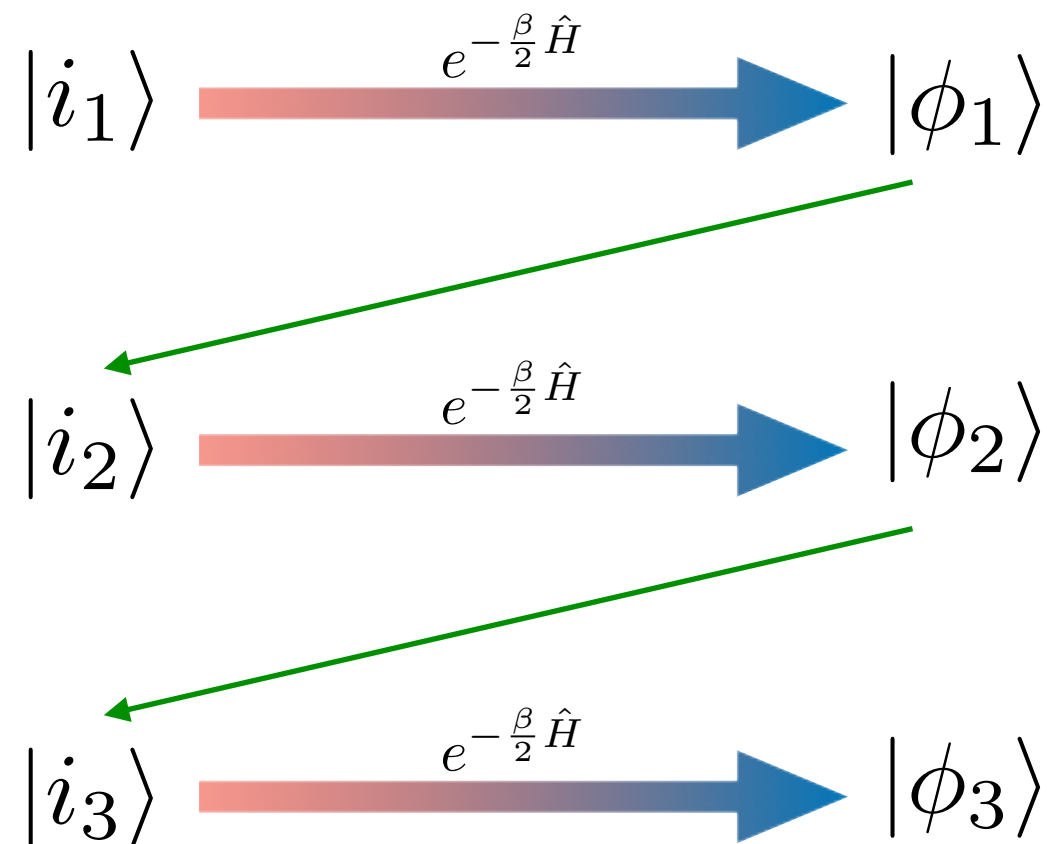
choose $|\phi_i\rangle \propto e^{-\frac{\beta}{2} \hat{H}} |i\rangle$ to "descend" from
untangled (zero-entanglement) states



- ☑ **Solved** problem of representing $|\phi_i\rangle \propto e^{-\frac{\beta}{2}\hat{H}}|i\rangle$
(choose $|i\rangle$ as product states)

One more **problem**: too many states –
there are exponentially many $|i\rangle$ and thus $|\phi_i\rangle$

- ☑ **Solution**: sample over the $|i\rangle$



$$p(\phi_1 \rightarrow i_2) = |\langle i_2 | \phi_1 \rangle|^2$$

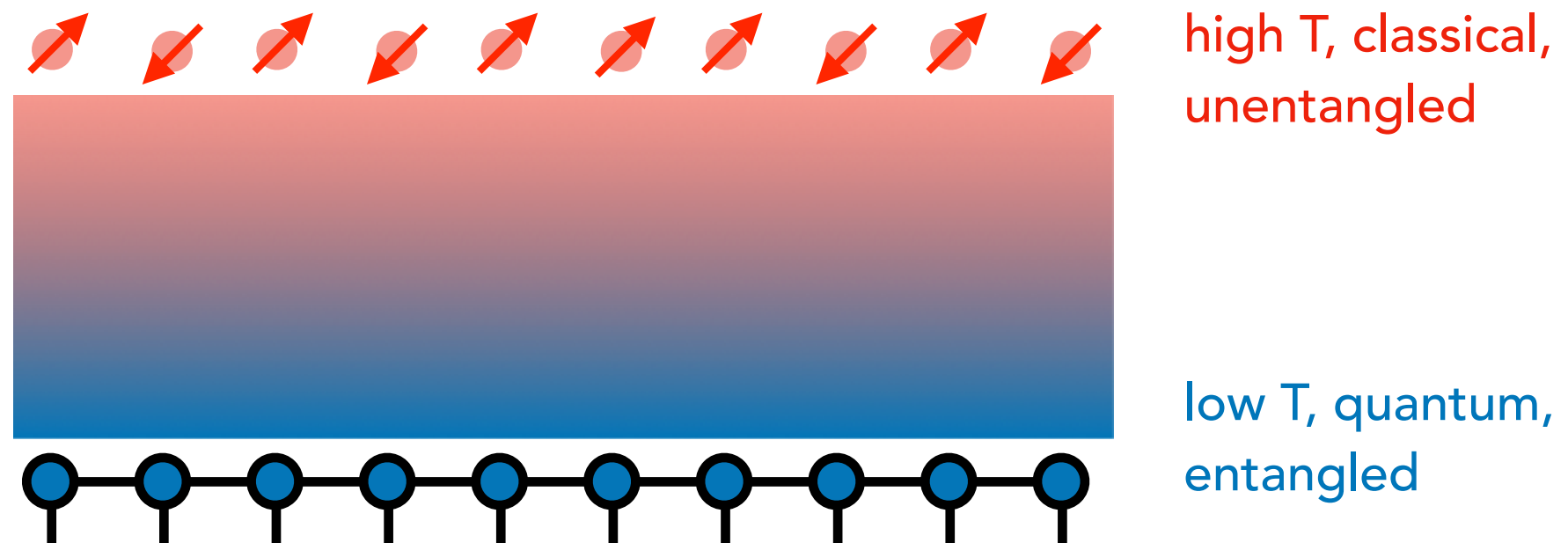
$$p(\phi_2 \rightarrow i_3) = |\langle i_3 | \phi_2 \rangle|^2$$

Algorithm just described named
minimally entangled typical thermal states (**METTS**)^{1,2}

$$|\phi_i\rangle \propto e^{-\frac{\beta}{2} \hat{H}} |i\rangle \quad \text{METTS wavefunction}$$

Quantum Monte Carlo where samples are
entangled wavefunctions, not classical configurations

Classicality of METTS depend on T

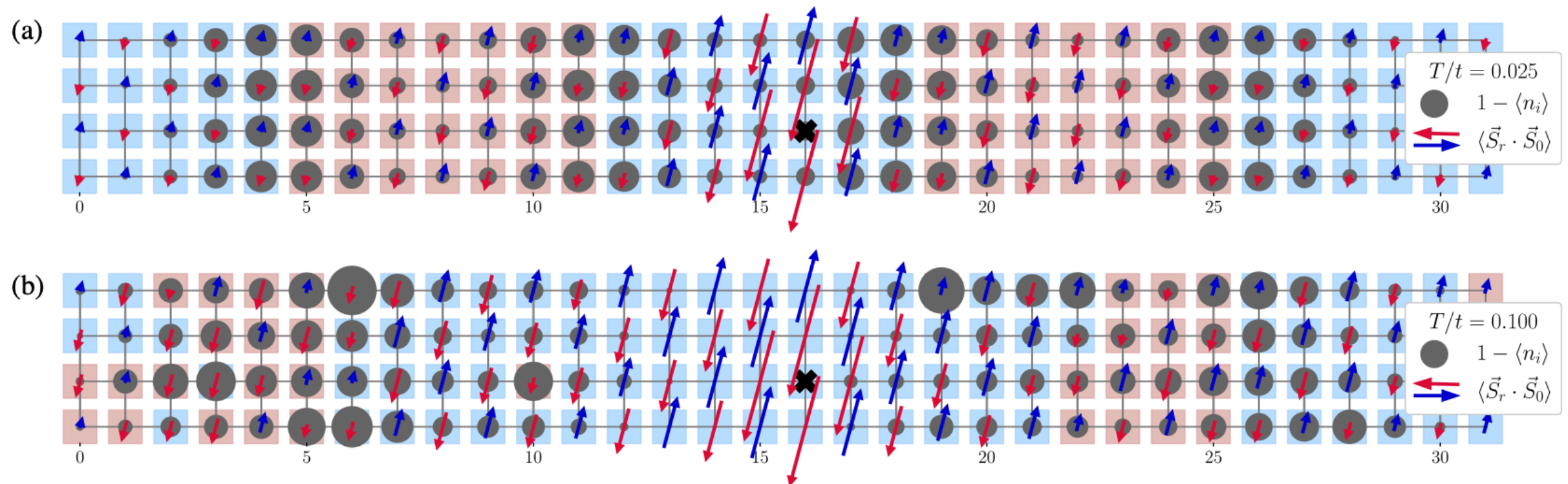


[1] S.R. White, PRL (2009)

[2] Stoudenmire, White, NJP (2010)

Minimally entangled typical thermal states (METTS)

Used by researchers here at CCQ to study finite temperature electronic systems [1]:



[1] A. Wietek et al, PRX 11 (2021)

Further Applications of TDVP and TEBD

- Open Quantum Systems $\rho(t) = \exp(\mathcal{L}t)\rho(0)$
- Green's functions
- OTOCS (Out-of-time ordered correlators)
- Many more...
- Now lets do some time evolution with [ITensorMPS.jl](#)